

Diagnostic accuracy of myocardial perfusion imaging and stress echocardiography for the diagnosis of left main and triple vessel coronary artery disease: a comparative meta-analysis

N Mahajan,¹ L Polavaram,¹ H Vankayala,¹ B Ference,² Y Wang,³ J Ager,³ J Kovach,² L Afonso²

¹Department of Internal Medicine, Detroit Medical Center, Wayne State University, Detroit, Michigan, USA

²Division of Cardiology, Detroit Medical Center, Wayne State University, Detroit, Michigan, USA

³Department of Biostatistics, Detroit Medical Center, Wayne State University, Detroit, Michigan, USA

Correspondence to

Luis Afonso, Internal Medicine Division of Cardiology, Wayne State University, Harper University Hospital, 8 Brush, 3990 John R, Detroit, Michigan 48201, USA; lafonso@med.wayne.edu

All the authors contributed significantly towards this manuscript.

Accepted 18 February 2010

ABSTRACT

Objectives Compare the diagnostic performance of stress echocardiography (SE) and myocardial perfusion imaging (MPI) for the diagnosis of left main disease (LM) and triple vessel disease (TVD).

Background Limited comparative data on MPI and SE for the detection of LM and TVD (high-risk coronary artery disease) exist in the literature.

Methods Quantitative meta-analysis was performed using studies identified by systematic electronic literature search. Articles were included if they reported data on exercise, dobutamine SE or exercise, adenosine, dipyridamole, thallium-201, technetium-99m sestamibi MPI and used coronary angiography as the reference test. Summary receiver-operating characteristic (SROC) curves were constructed for each imaging modality. Additionally, pooled sensitivity, specificity and likelihood ratios were calculated per modality. Meta-regression was performed to adjust for patient and study characteristics.

Results Thirty-two studies met inclusion criteria; 23 (MPI-15; SE-14; Common studies-6) provided sufficient data for analysis. In a SROC model comparing the two imaging modalities, SE was associated with higher area under curve (0.82 (0.03) vs 0.73 (0.02), $p=0.01$) and index Q* value (0.75 (0.02) vs 0.67 (0.02), $p=0.01$). Using pooled summary point estimates, SE had higher pooled sensitivity (94% vs 75%, $p<0.001$) and lower negative likelihood ratio (0.21 vs 0.47, $p<0.001$) compared to MPI. No evidence of a difference in the pooled specificity (40% vs 48%, $p=0.16$) and positive likelihood ratio (1.52 vs 1.58, $p=0.36$) was seen between the two stress modalities. Pooled diagnostic OR on meta-regression (9.78 vs 4.06, $p=0.02$) remained significantly higher for SE compared to MPI for identification of LM and TVD even after adjustment for study characteristics.

Conclusions Since LM alone or in combination with TVD are categorised as representing potentially life-threatening variants of CAD, a screening test with high sensitivity, low negative likelihood ratio or higher discriminatory capacity would be desirable for risk stratification. In the absence of a direct head-to-head comparison of the diagnostic accuracies of SE and MPI, our findings indicate that SE appears to be the preferred screening modality for high-risk coronary artery disease.

nary artery disease (CAD). Stress echocardiography (SE) and myocardial perfusion imaging (MPI) facilitate non-invasive risk stratification and identification of patients with significant CAD warranting coronary angiography or revascularisation.^{1–6}

Traditionally, invasive coronary angiography has served as the gold standard for characterising the diagnostic accuracy of non-invasive testing, generally expressed in terms of sensitivity (percentage of true-positive tests in those with angiographic CAD) and specificity (percentage of true negative tests in those without angiographic CAD).

Published values for the sensitivity and specificity of existing stress modalities for the overall detection of CAD suggest comparable accuracy between MPI and SE.^{3–8} A pooled analysis of 18 studies involving patients who underwent exercise or pharmacologic stress in conjunction with thallium- or technetium-labeled radioisotope imaging for the diagnosis of CAD, reported sensitivity and specificity of 80% and 86% for SE and 84% and 77% for MPI, respectively.⁹

Left main (LM) involvement alone or in combination with additional coronary involvement, LM equivalent (proximal left anterior descending and proximal circumflex disease) and triple vessel disease (TVD) represent high-risk subsets of obstructive CAD (high-risk CAD). LM disease defined as a >50% stenosis of this vessel is found in approximately 7% of patients undergoing coronary angiography¹⁰ and is associated with significant cardiac mortality.^{11–12} Similarly, 5-year survival with medical therapy in patients with TVD ranges between 79% to as low as 59%, when the proximal left anterior descending artery is involved.¹³ Revascularisation in LM^{14–16} and TVD¹⁷ has been shown to positively impact survival.

While the available literature on MPI and SE suggest similar performance between these modalities for the overall detection of CAD,^{3–8 18 19} data on the comparative performance of SE versus MPI for the detection of “high-risk CAD” are very limited.^{4 6 7} Specifically, no conclusive evidence as to whether MPI or SE is more vulnerable to false negative results (low sensitivity) among persons with LM and TVD CAD exists in the literature. This meta-analysis was undertaken to compare the diagnostic performance of SE and MPI against the conventional gold standard of coronary angiography, using existing data on patients with LM and TVD.

Non-invasive testing is routinely employed for the detection of clinically suspected obstructive coro-

METHODS

Data extraction

A systematic electronic literature search of OVID, EMBASE, the Cochrane database and MEDLINE using PubMed as the search engine was used to identify studies reporting comparative data on SE and MPI for LM and TVD (1980–2007 in the English language). The following keywords were used for the search: “stress echocardiography, dobutamine, dipyridamole, adenosine, technetium-99m sestamibi, technetium-99m tetrofosmin and thallium-201 myocardial perfusion imaging”, single photon emission computerised tomography–myocardial perfusion imaging (SPECT-MPI) “myocardial perfusion scintigraphy”, “left main disease”, “triple vessel disease”, “coronary artery disease”, “diagnostic accuracy” and “comparative study”. Articles were also identified using the “related articles” function in PubMed. Additional references were obtained from bibliographies of original papers and review articles.

Eligibility criteria and data extraction

All studies comparing SE or MPI to conventional angiography and reporting data on LM or TVD were considered. Studies where patients did not undergo angiography and studies that did not report on the numbers of patients diagnosed with LM or TVD were excluded. Only studies that provided angiographic evidence of LM and TVD were included. Non-invasive imaging tests included exercise or pharmacologic SE and exercise or pharmacologic MPI. Authors of these studies were not contacted directly. Data were extracted on author, journal, date of publication, institution, study design, patient demographics (mean age, percentage of male patients) and technical aspects of the studies.

All data were extracted independently by three reviewers (NM, YW and JA), and any discrepancy was resolved by consensus. Sensitivity and specificity were required for analysis, and accordingly, studies providing insufficient information for calculations were excluded. Data were extracted using any “reversible perfusion deficit (PD)”, “electrocardiographic changes in ≥ 2 contiguous leads”, “transient ischemic dilatation” or “hypotension” or “other ancillary data” in studies with MPI and any “reversible regional wall motion abnormality” for SE as reported by the authors. Fixed perfusion abnormalities and wall motion abnormalities at rest were excluded during extraction of data for positive results to avoid any impact of prior myocardial infarction. We also extracted data for high-risk features including “LM” or “multivessel (≥ 2 coronary vessel)” scintigraphic pattern in studies with MPI and “reversible regional wall motion in multivessel territory (≥ 2 coronary vessel)” abnormalities for SE (stringent criterion) in order to define their diagnostic accuracy for precise identification of multivessel CAD.

Outcome measures and definitions

The primary outcome measure was the construction of summary receiver-operating characteristic (SROC) curves for both the imaging modalities. Summary receiver analysis quantified MPI or SE performance using area under the curve (AUC), diagnostic OR (DOR) and index Q^* value as summary estimates.^{20–22} The SROC curve was described for use in the assessment of diagnostic tests in 1971 and reflects the discriminating ability of a diagnostic test. It can overcome some of the limitations associated with pooling sensitivities and specificities of published studies; accordingly, it has become the preferred method to assess performance of a diagnostic test based on data from a meta-analysis.²³ Area under the SROC curve reflects overall performance of the test and is robust to heterogeneity. It

is expressed as a value between 0 and 1, with higher values indicating better test performance. An area of 1.0 is representative of a perfect test, while an area of 0.5 represents a test whose performance is no different from a random guess. False positive rate (FPR=1–specificity) is the x axis, while true positive rate (TPR=sensitivity) is the y axis of SROC curve. Index Q^* represents the value of sensitivity and specificity implied by the SROC curve at the point where the two are equal. The closer it is to the top left corner, the better the accuracy (higher sensitivity and specificity). It is invariant to heterogeneity.²²

The individual study sensitivities and specificities, diagnostic odds ratio (DOR) and likelihood (positive and negative) ratios of the two imaging modalities (for high-risk CAD) were extracted using two-by-two contingency tables for each end-point of correct patient diagnosis. This was either quoted directly in the studies or extracted from the analysis of the true positives (TP), true negatives (TN), false positives (FP) and false negatives (FN) on a per patient basis. Data per lesion were published in the included studies for MPI using scintigraphic perfusion abnormality along with ancillary data and visual assessment of regional left ventricular wall thickening/motion abnormalities in SE studies. The study defined all persons without either LM or TVD CAD as “not diseased” for the purposes of analysis. The accuracy based on index vessel stenosis severity was not estimated, as these data were not consistently available in the literature for SE and MPI.

Since patients with one- and two-vessel coronary artery disease were classified as “not diseased”, specificity for both modalities was artificially lowered.

We also computed likelihood ratios (LR) as they provide a measure of a test’s ability to “rule in” or “rule out” disease independent of disease probability. The positive LR represents the OR of a positive test result being obtained in a diseased patient population compared to non-diseased patients. A test with higher positive LR is useful in confirming the presence of disease. The negative LR represents the OR that a negative test result will be observed in a diseased population compared to the odds that the same result will be observed among a non-diseased population. A test with negative LR closer to 0 helps “rule out” disease. They are computed as:

$$\text{Positive LR} = \frac{\text{Sensitivity}}{1 - \text{Specificity}}$$

$$\text{Negative LR} = \frac{1 - \text{Sensitivity}}{\text{Specificity}}$$

An ideal screening test has high sensitivity and low negative LR.

Since sensitivity and specificity are only partly representative of a test’s discriminatory performance (ignoring threshold differences), DOR is widely utilised as a single indicator of test performance and is independent of the prevalence of disease.²⁴ It was computed for the included studies as follows:

$$\text{DOR} = \frac{\text{TP} \times \text{TN}}{\text{FP} \times \text{FN}} = \frac{(1 - \text{False positive rate}) / (\text{False positive rate})}{(1 - \text{True positive rate}) / (\text{True positive rate})}$$

False positive rate is the x axis, while true positive rate is the y axis of the SROC curve. The value of a DOR can range from 0 to

Cardiac imaging and non-invasive testing

infinity. Higher DOR values are reflective of better discriminatory performance of a diagnostic test. Pooled DOR can be derived from SROC curves.²⁴

A weighted meta-regression (weighted by the inverse of variance of DOR in each study) was also performed for both imaging modalities to evaluate the importance of potential effect modifiers to explain variation between studies.²⁵ We assessed the influence of variables including publication year, mean age, proportion of men in the study population, clinical indication, type of stressor (exercise or pharmacologic) and year of publication.

Heterogeneity among the included studies was assessed using the Cochrane Q test. This test generates a probability (p value), based on a χ^2 distribution, that between study differences in results equal to or greater than those observed are likely to occur simply by chance ($p < 0.1$ representing significant statistical heterogeneity). The I^2 statistic (measure of proportion of inconsistency not explained by chance) is calculated from Cochrane Q and is used to quantify heterogeneity. It can vary from 0% to 100%, with a value of 0% representative of absence of any observed heterogeneity ($I^2 > 50\%$ represents substantial heterogeneity); τ^2 gives an estimate of between study variance.²⁶

Statistical analysis

SROC curves were constructed for the two imaging modalities and analysed to compare area under the curves, index Q^* values and DOR to assess their discriminatory capacity for detection of high-risk CAD. A symmetric SROC model was used to address the lack of numerical convergence of the full SROC model. Pooled sensitivity, specificity, positive and negative likelihood ratios (weighted by the inverse of variance of sensitivity, specificity and likelihood ratios in each study, respectively), with 95% CI, were calculated using a random effects model.²⁷ Heterogeneity was explored using sub-group analyses and meta-regression. Study-specific DORs were meta-regressed to test the effect of potential effect modifiers.²⁵ We also calculated the effect of varying the positivity criteria using "LM" or "multivessel" scintigraphic pattern in studies with MPI and based on "reversible regional wall motion in multivessel territories" abnormalities for SE (stringent criterion). Sub-group analyses were performed for both the imaging modalities. Analysis was conducted using SPSS V.15.0 (SPSS Inc.), Windows STATA V.10.0 (StataCorp LP), StatsDirect (StatsDirect Ltd) and Meta-Test Software V.0.9 (Tufts, New England Medical Centre). The Egger's precision-weighted linear regression method was used to statistically assess publication bias ($p < 0.05$ was considered representative of significant statistical publication bias).²⁸ This study was undertaken in accordance with previously reported guidelines for meta-analyses evaluating diagnostic tests.^{20–22 29 30}

RESULTS

Study eligibility

Thirty-two studies matched the selection and search criteria and were obtained for evaluation in full.^{31–62}

High-risk CAD (LM+TVD)

Twenty articles utilised MPI for LM and TVD detection.^{33 35–37 39 40 42–47 49 50 56–61} Five of these studies were excluded due to lack of extractable data,^{42 59} and inability to abstract specificity as the studies included only patients with LM or TVD.^{33 58 61} Eighteen studies with comparative data on SE for LM and TVD were identified.^{31 32 34 36–41 47–49 51–55 62} Three studies using dipyridamole^{32 53 54} stress echocardiography (as this stress modality is not commonly employed in the US) and trans-

esophageal echocardiography⁵⁵ were excluded. Thus, we were left with 15 studies for MPI (exercise-9, adenosine-1, dipyridamole-1, dobutamine-3, both exercise and pharmacologic-1) and 14 studies for SE (exercise-5, dobutamine-9). Six studies reported use of both MPI and SE to diagnose CAD in the same cohort.^{36 37 39 40 47 49} The study characteristics of the included studies for high-risk CAD analysis are shown in table 1. Data on high-risk CAD patients with reduced left ventricular ejection fraction were not uniformly abstractable for the included studies.

Statistical analysis for MPI studies

A total of 2310 patients underwent MPI, ranging from 70 to 466 participants in the 15 studies, were included for analysis. The prevalence of high-risk CAD was 28% (649/2310) from the included studies. The SROC analysis for all the included studies yielded overall weighted AUC for high-risk CAD of 0.73 (0.02) with index Q^* value of 0.67 (0.02) (figure 1). Using pooled summary point estimates, sensitivity, specificity, positive and negative likelihood ratios for MPI were 75% (72–78%), 48% (35–68%), 1.58 (1.33–1.74) and 0.47 (0.39–0.56) respectively. The forest plot for sensitivity (figure 2) showed heterogeneity (Cochrane Q $p < 0.1$). Six small-sample-size studies were primarily responsible for this heterogeneity.^{37 39 43 45 47 49} However, there was no significant heterogeneity in the forest plot for negative likelihood ratio (1–Sensitivity/Specificity; figure 3), implying that it is reasonable to combine the included studies to obtain pooled estimates. The pooled DOR for high-risk CAD was 4.06 (2.84–5.80) and supported absence of any substantial heterogeneity with Cochrane Q of 19.93 ($p = 0.13$) with $I^2 = 29.7\%$ (figure 4).

Statistical analysis for SE studies

A total of 1403 patients underwent SE, with a range of 37–309 participants in the 14 studies included for analysis. The prevalence of high-risk CAD was 18% (254/1403) from the included studies. The SROC analysis for all the included studies yielded overall weighted AUC for high-risk CAD of 0.82 (0.03) with index Q^* value of 0.75 (0.02) (figure 1). Using summary point estimates, sensitivity, specificity, positive and negative likelihood ratios for SE were 94% (91–97%), 40% (37–43%), 1.52 (1.38–1.68) and 0.21 (0.13–0.33), respectively. Lack of significant heterogeneity seen in the forest plots for sensitivity (figure 2) and negative likelihood ratios (figure 3), as evidenced by Cochrane Q $p > 0.1$, permitted the combination of included studies to obtain pooled estimates. The pooled DOR for high-risk CAD was 9.78 (5.83–16.40; figure 4). The included studies showed no evidence of heterogeneity with Cochrane Q of 8.22 ($p = 0.83$) and absence of any inconsistent studies ($I^2 = 0$ attributing all the variability in estimates to sampling error within included studies) or between-study variance ($\tau^2 = 0$).

Results of indirect comparison between MPI and SE

Our results suggest significantly higher area under SROC curve (0.73 (0.02) vs 0.82 (0.03), $p = 0.01$) and index Q^* value (0.75 (0.02) vs 0.67 (0.02), $p = 0.01$) for SE supporting its overall superiority in detection of high-risk CAD. Additionally, SE had higher pooled sensitivity (94% vs 75%, $p < 0.001$) and DOR (9.78 vs 4.06, $p = 0.02$), and lower negative likelihood ratio (0.21 vs 0.47, $p < 0.001$). No evidence of a difference in the pooled specificity and positive likelihood ratios was seen for the two imaging modalities in detection of high-risk CAD (table 2). Exclusion of heterogeneous studies yielded higher sensitivity for SE (94% vs 75%, $p < 0.01$) without any evidence of a difference in

Table 1 Summary point estimates for high-risk CAD (LM and TVD) analysis

ID	Year	Author	Method	Age (years)	Male (%)	Disease†	Total Sample	Positivity Criterion	TP	FN	TN	FP
1	1981	Rehn ⁵⁰	MPI	51	88	45	104	Reversible PD	37	8	21	38
2	1984	Nygaard ⁴⁶	MPI	52	81	96	295	Reversible PD or hypotension or ECG changes	72	24	99	100
3	1986	Maddahi ⁴⁴	MPI	56	81	56	105	Reversible PD or hypotension or ECG changes	34	22	41	8
4	1986	Wakasugi ⁵⁷	MPI	NA	NA	56	79	Reversible PD	35	21	23	0
5	1990	Mahmorian ⁴⁵	MPI	56	74	13	296	Reversible PD	13	0	86	197
6	1991	Chikamori ³⁵	MPI	61	78	133	466	Reversible PD or hypotension or ECG changes	76	57	251	82
7	1992	Quinones ⁴⁹	MPI	57	67	16	72	Reversible PD	15	1	41	15
8	1993	Solot ⁴³	MPI	61	62	20	127	Reversible PD	20	0	29	78
9	1993	Hecht ⁴⁰	MPI	58	86	15	71	Reversible PD	10	5	17	39
10	1994	Takeishi ⁵⁶	MPI	63	63	11	75	Reversible PD or abnormal L/H ratio or LV dilation	9	2	42	22
11	1997	Oguzhan ⁴⁷	MPI	51	84	14	70	Reversible PD	14	0	17	39
12	1998	Elhendy ³⁶	MPI	60	63	15	84	Reversible PD	11	4	33	36
13	1998	Elhendy ³⁷	MPI	58	0	5	70	Reversible PD	5	0	34	31
14	1999	Fragasso ³⁹	MPI	61	54	11	101	Reversible PD	11	0	17	73
15	2003	Lima ⁶⁰	MPI	61	65	143	112	Reversible PD or wall motion or ECG or T1D or LV enlargement or reduced LVEF	126	17	38	74
16	1991	Sawada ⁵²	SE	59	62	7	55	New or worse RWMA	7	0	21	27
17	1992	Epstein ³⁸	SE	59	70	7	61	New or worse RWMA	7	0	11	43
18	1992	Quinones ⁴⁹	SE	57	67	16	112	New or worse RWMA	15	1	44	52
19	1993	Hecht ⁴⁰	SE	58	86	15	71	New or worse RWMA	14	1	23	33
20	1993	Ryan ⁵¹	SE	57	75	55	309	New or worse RWMA	53	2	92	162
21	1994	Prince ⁴⁸	SE	62	86	9	73	New or worse RWMA	8	1	50	14
22	1994	Sharp ³¹	SE	58	NA	9	54	New or worse RWMA	9	0	9	36
23	1995	Bjornstad ³⁴	SE	58	81	6	37	New or worse RWMA	5	1	8	23
24	1997	Oguzhan ⁴⁷	SE	51	84	14	70	New or worse RWMA	14	0	24	32
25	1997	Hennesy ⁴¹	SE	59	64	67	221	New or worse RWMA	65	2	55	99
26	1997	Rallidis ⁶²	SE	57	NA	18	85	New or worse RWMA	16	2	21	46
27	1998	Elhendy ³⁶	SE	60	63	15	84	New or worse RWMA	12	3	30	39
28	1998	Elhendy ³⁷	SE	58	0	5	70	New or worse RWMA	5	0	33	32
29	1999	Fragasso ³⁹	SE	61	54	11	101	New or worse RWMA	10	1	41	49

The TPR or TNR have to be modified when reported TPR or TNR=1 since 0 cannot be in the 2×2 table. We add 0.5 to each cell of the 2×2 table if any 0 exists as recommended by Walter.²¹

TP, true positive; FN, false negative; TN, true negative; FP, false positive; MPI, myocardial perfusion imaging; SE, stress echocardiography.

†Disease-cumulative number of patients with left main and triple vessel disease.

Cardiac imaging and non-invasive testing

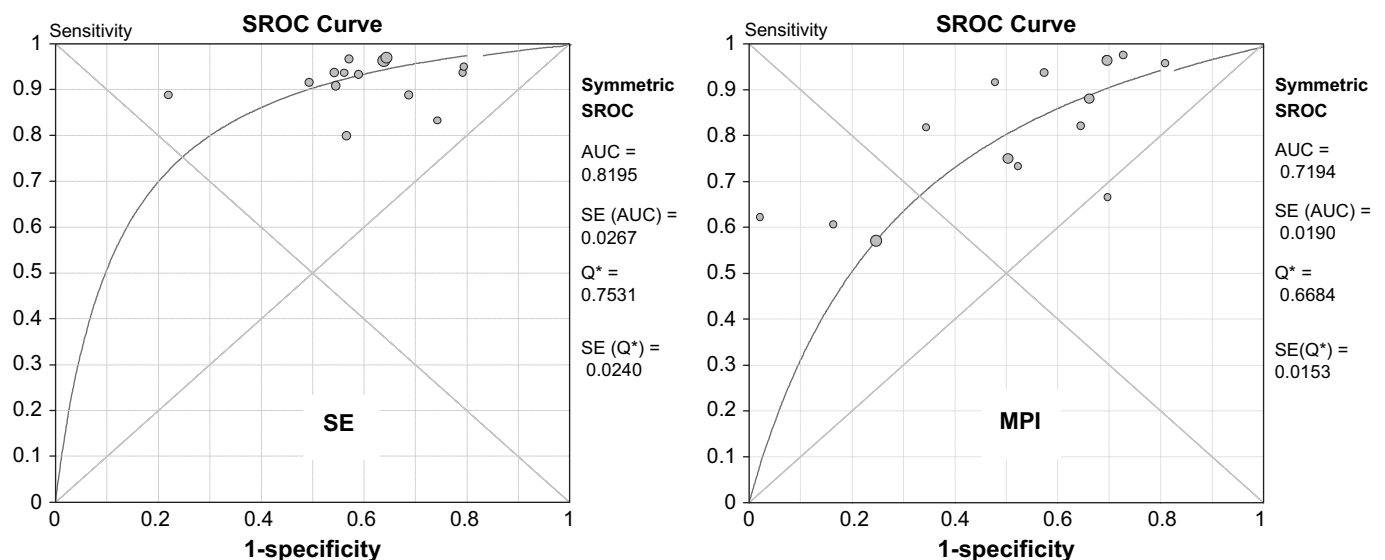


Figure 1 SROC comparison for SE and MPI for detection of high-risk CAD (LM or TVD). The points are shrunk estimates taken from the fitted random effects model, and may not correspond to the sensitivities and specificities apparent from table 1. (SROC, summary receiver operating characteristic curves; AUC, area under curves; SE, stress echocardiography; Q*, Index Q* value).

specificity (43% vs 62%, $p=0.14$). In univariate analysis for MPI and SE, mean age, sex (proportion of men) and year of publication were significant ($p<0.05$) predictors, but clinical indication and type of stressor (exercise vs pharmacologic) were not. In multivariate analysis, the difference in DORs remained significant even after adjustment for mean age, sex (proportion of men) and year of publication. Analysis of data extracted using “LM” or “multivessel” scintigraphic pattern in studies with MPI and based on “reversible regional wall motion abnormality (RWMA) in multivessel distribution” (stringent criterion) supported the same conclusion. However, the sensitivity for detecting high-risk CAD decreased while the specificity increased using stringent criterion for both the modalities in comparison to data reported in table 2. No evidence of publication bias was found by using Egger’s precision-weighted linear regression. We also performed separate sub-group analyses to investigate the heterogeneity.

Sub-group analysis

Several sub-group analyses were performed for the two stress imaging modalities:

1. Comparison of the performance of SE and MPI in detecting high-risk CAD in terms of reversible PD (for MPI) and reversible RWMA (for SE)

For detection of high-risk CAD, SE had a significantly higher area under SROC curve (0.82 vs 0.73, $p=0.03$), pooled DOR (9.78 vs 3.51, $p=0.02$) and sensitivity (94% vs 66%, $p<0.001$) for detection of high-risk CAD. However, no difference in specificity was observed (0.40 vs 0.53, $p=0.26$).

2. Comparison of the performance of SE and SPECT-MPI for detection of high-risk CAD after exclusion of MPI studies utilising planar scintigraphy.

This analysis included 3130 patients (SPECT-1727; SE-1403). The results were similar even after exclusion of planar imaging studies.^{44 46 50 57} SE had significantly higher area under SROC curves (0.82 vs 0.73, $p=0.02$) with superior sensitivity (94% vs 78%, $p<0.01$), negative likelihood ratio (0.21 vs 0.44, $p<0.01$) and DOR (9.78 vs 3.93, $p=0.02$). However, paucity of abstractable data on studies incorporating recent advances (including gated-SPECT, spatial normalisation and novel echocardiography technologies such as myocardial contrast perfusion, transesophageal,

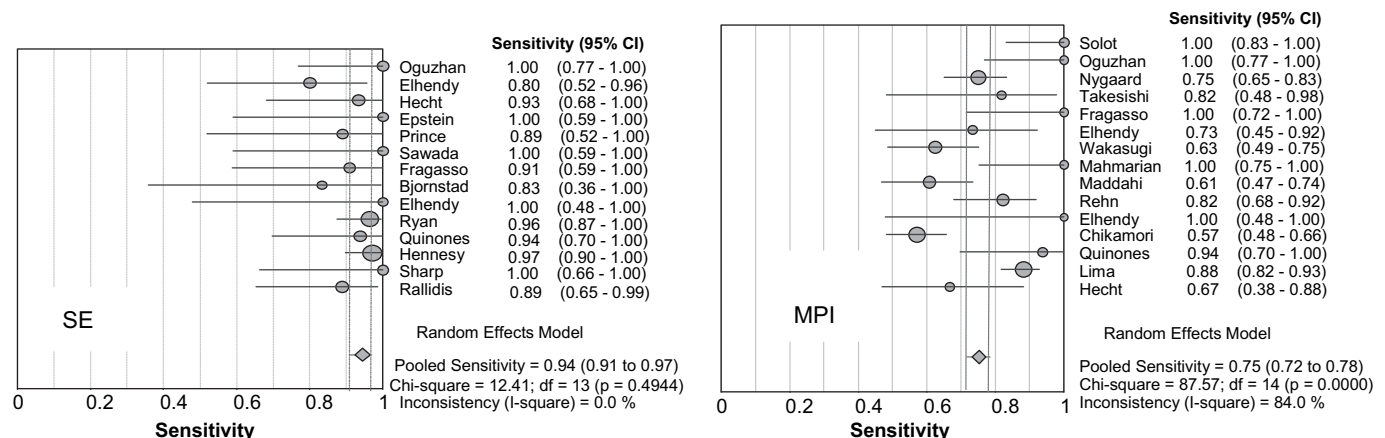


Figure 2 Comparison of sensitivity of SE and MPI for detection of high-risk CAD.

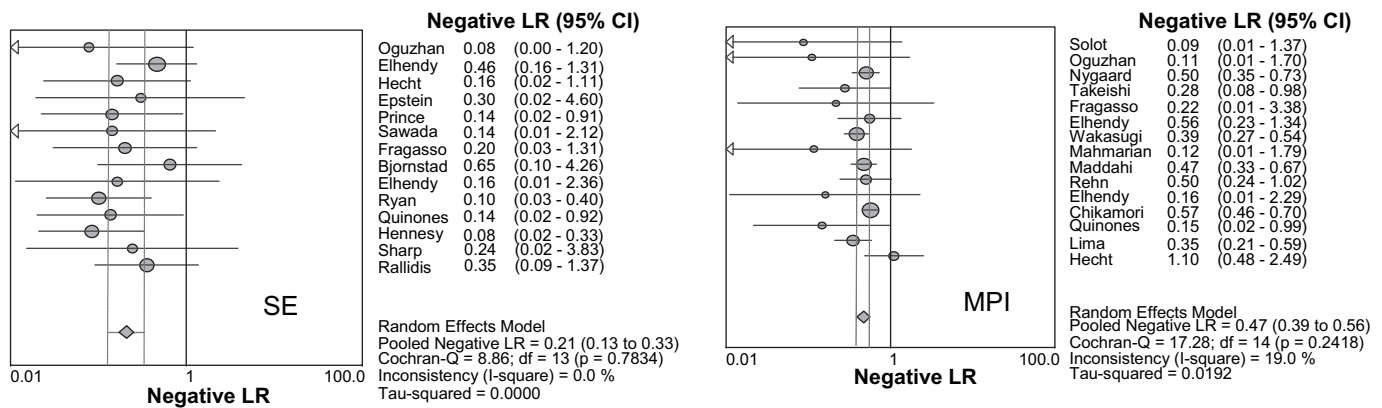


Figure 3 Comparison of negative likelihood ratios of SE and MPI for detection of high-risk CAD.

harmonic and strain imaging) precluded any meaningful contemporary comparisons.

3. Comparison of the performance of SE and MPI for detection of high-risk CAD using data from studies with reported outcome data for both stress modalities by the same author^{36 37 39 40 47 49}

The sample size was significantly lower for this analysis and added up to roughly one-sixth of patients with high-risk CAD. SE had a significantly higher area under SROC curve (0.80 vs 0.70, $p=0.04$) and higher sensitivity (93% vs 84%, $p=0.04$) without any evidence of a difference in specificity (46% vs 39 %, $p=0.14$) and DOR (7.00 vs 4.47, $p=0.32$) in comparison to MPI.

4. Comparison of the performance of SE and MPI for detection of high-risk CAD excluding data from studies with reported outcome data for both stress modalities by the same author^{31 32 34 35 38 40 41 43-46 48 50-52 56 57 60 62}

This analysis included 2737 patients. Even after excluding the six common studies that contributed to both stress modalities, the conclusion (better discriminatory ability of SE) remained unchanged with a significantly higher area under SROC curves (0.83 vs 0.73, $p=0.01$), sensitivity (96% vs 74%, $p<0.001$) and DOR (9.05 vs 4.30, $p=0.01$).

5. Comparison of the performance of SE and MPI for detection of high-risk CAD using studies categorised by stressor modality (exercise or pharmacologic)

Regardless of type of stressor used, comparison of studies involving SE and MPI using exercise or pharmacological stressor yielded results consistent with the overall trend observed (study where data on exercise and pharmacologic

stressor was not abstractable (60) was excluded from this analysis). The sensitivity of SE remained significantly higher compared to MPI. (Exercise—76% vs 95%, $p<0.01$; Pharmacologic—62% vs 93%, $p<0.01$).

6. Comparison of the performance of SE and MPI for detection of Left Main disease

The absence of adequate number of studies for SE ($n=3$) and MPI ($n=6$) reporting abstractable data on left main disease detection precluded a meaningful analysis and therefore was not pursued further.

These sub-group analyses appear to support the superior ability of SE for detection of high-risk CAD.

DISCUSSION

The results of our meta-analysis suggest that SE has a better discriminatory capacity (on SROC analysis) than MPI for the detection of high-risk CAD. This difference remains significant even after adjustment of sex, mean age of sample population and year of publication. Both tests showed high AUC values (table 2) with high DORs (figure 4). However, SE appears to have a higher sensitivity and lower negative likelihood ratio than MPI when performed in persons with LM or TVD. These findings suggest that SE is less vulnerable than MPI to false negative tests among persons with high-risk CAD. Because persons with high-risk CAD may derive mortality benefit from revascularisation, this finding may have important clinical implications. Importantly, this is the first comparative analysis of two routinely employed stress-imaging modalities in the assessment

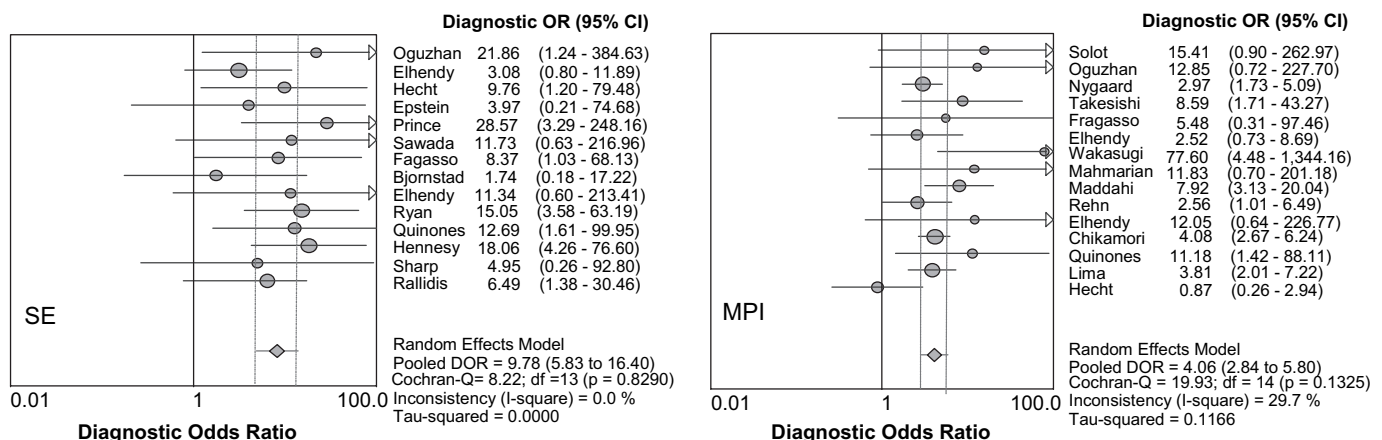


Figure 4 Comparison of DORs for studies using SE or MPI for detection of high-risk CAD.

Cardiac imaging and non-invasive testing

Table 2 Individual study results for diagnosis of coronary artery disease comparing MPI and SE with conventional angiography

Coronary vessel	Sensitivity			Specificity			SROC AUC [†] (SE [†])			Meta Regression DOR		
	MPI	SE	p Value	MPI	SE	p Value	MPI	SE	p Value	MPI	SE	p Value
High-risk CAD (LM and TVD)	0.75 (0.72–0.78)	0.94 (0.91–0.97)	<0.001	0.48 (0.35–0.68)	0.40 (0.37–0.43)	0.16	0.73 (0.02)	0.82 (0.03)	0.01	4.06 (2.84–5.80)	9.78 (5.83–16.40)	0.02

SROC, summary receiver operating characteristic curve; AUC, area under curve; SE, standard error; DOR, diagnostic odds ratio.

of high-risk CAD. The finding that SE has a higher discriminatory capacity compared to MPI for the detection of high-risk CAD was consistently observed both in pooled analyses of studies that employed both SE and MPI in each study participant and in pooled analyses of studies that employed only one or the other modality to image each study participant.

Overall comparative data between imaging modalities

Both SE and MPI have proven to be comparable and extremely useful in the overall evaluation of CAD.^{4 7 8 63 64} Several studies comparing diagnostic accuracy of SE with MPI in the same population have shown that although both tests have comparable sensitivity for the diagnosis of CAD, SE has higher specificity.^{1–5} In one study that compared dobutamine–atropine SE to dipyridamole–sestamibi perfusion imaging in the same population, both modalities had similar sensitivities; however, dobutamine–atropine SE demonstrated a higher specificity (95% vs 76%).⁵ Similarly, a meta-analysis of 17 studies reported a marginal decrease in sensitivity for SE compared to MPI (80% vs 84%, $p<0.05$) but a higher specificity (77% vs 86%, $p<0.001$) for the overall detection of CAD.⁹ This may be ascribed to the fact that MPI assesses relative and regional differences in myocardial perfusion, the initial event in the ischemic cascade,⁶⁵ which precedes the development of contractile abnormalities (the basis of SE).

Underestimation of high-risk CAD by MPI

It is important to recognise that the favourable overall MPI diagnostic accuracy data may not be extrapolated to high-risk CAD sub-groups. Several factors may contribute to the underestimation of high-risk CAD and include inadvertent or unrecognised ingestion of caffeine containing products or aminophylline (antidotes to adenosine or dipyridamole), anti-angina therapy, methodological artefacts and attenuation from motion, soft tissue or diaphragmatic attenuation and resolution issues partly related to collimation.⁶⁶

Plateauing of tracer uptake at high flow rates, leading to masking of disparities in coronary flow reserve, particularly if vasodilator stress employed has also been implicated. Indeed, the phenomenon of balanced ischaemia is perhaps the most vulnerable aspect of MPI, because detecting abnormal myocardial perfusion, hinges on demonstrating relative disparities in perfusion between vascular territories. Specifically, in the setting of multivessel disease, the uniform reduction in flow (lack of regional heterogeneity in myocardial perfusion), coupled with the inability to measure absolute flow, may preclude delineation of ischemic areas^{66 67} in contradistinction to PET imaging that permits flow quantification or stress echocardiography that relies on the demonstration of regional contractile abnormalities. This limitation was borne out in a recent study by Ragosta *et al*, wherein 36 patients with previous MPI underwent coronary angiography with fractional flow reserve (FFR) estimation (88 diseased segments that included a total of 20 chronic total occlusions were studied). These investigators found that 36% of vascular zones

with chronically occluded arteries or a low fractional reserve (FFR<0.75) demonstrated no perfusion defects.⁶⁸

The limitations of MPI for the detection of significant LM stenosis are not well understood. The intuitively expected “LM pattern” is apparent in only a minority of patients with LM stenosis^{33 61} and is observed in only 5–13% of patients with angiographically documented LM disease.^{50 57} It has been hypothesised that the inferior wall (subtended by the right coronary artery), which is typically subject to soft tissue diaphragmatic attenuation in men, conceivably contributes to an apparent balanced reduction pattern and might explain some of the false negative results in patients with LM disease.³⁷

It appears that gated SPECT-MPI with visual or quantitative assessment also underestimates extent of ischaemia in patients with high-risk CAD.^{33 60} A recent study by Berman *et al* reported findings on a subset of 101 patients (derived from a database of 1864 patients) who were identified to have significant angiographic LM disease requiring referral for revascularisation who underwent gated-SPECT imaging. Only 56% of patients were identified as having high-risk disease (>10% myocardium at stress) based on perfusion imaging, while 13% had normal perfusion by visual analysis.³³ With the exception of LM with TVD, <60% of patients with LM disease associated with other angiographic subtypes (in combination with either right coronary artery or left anterior descending or circumflex arteries) demonstrated multivessel perfusion defects or >10% myocardium-stress perfusion defects.^{33 61} Similarly, functional assessment has been shown to be of incremental value over perfusion alone in identification of triple vessel disease.⁶⁰ The integration of ancillary findings such as abnormal tracer wash out, hemodynamic response to rest and stress following adenosine,⁶⁹ low ejection fraction, attenuation correction,⁷⁰ abnormally increased lung uptake of radioisotope,⁷¹ RWMA and transient ischemic dilation (TID)^{32 34 37 39 48} have been shown to vastly improve the sensitivity for detection of LM disease (Berman *et al*³³ 83%, Maddahi *et al*⁴⁴ 77%, Chikamori *et al*³⁵ 72%), particularly LM in conjunction with TVD.

Although TID is an accepted marker of severe and extensive CAD and poor prognosis, Emmett *et al* demonstrated that TID is less frequently associated with high-risk CAD in patients with diabetes or left ventricular hypertrophy.⁷² The detection of high-risk CAD is further confounded by the fact that finding one abnormal coronary territory (either reversible or fixed defect) on perfusion imaging does not necessarily confer a benign prognosis and may be associated with TVD or LM in up to 26% of patients undergoing MPI.⁷³

Indeed, despite the incorporation of non-perfusion parameters, a number (up to 17%) of patients with high risk,^{33 61} potentially life-threatening CAD are missed by MPI (both visual and quantitative) or have minor/equivocal defects that would classify them as being low risk, obviating the need for further angiographic correlation or follow-up.

Use of right ventricular/left ventricular perfusion index has been advocated to circumvent the limitation of “balanced perfusion” encountered with SPECT in patients with high-risk

CAD.⁵⁹ Additionally, appropriate spatial normalisation, gating and the use of three-dimensional SPECT MPI have been recommended to improve detection of severe CAD.^{74–75} However, regrettably, these strategies remain widely under-utilised in the “real world” of contemporary nuclear cardiology practice, worldwide. Use of hybrid imaging (such as ⁸² Rubidium PET/CT imaging) technology has been found to provide incremental diagnostic information compared to MPI alone (permitting clinicians to delineate coronary anatomy, quantitate absolute blood flow, ejection fraction reserve and assess physiologic significance of flow limiting stenosis).^{76–77} It is hoped that hybrid imaging will accurately characterise patients with high-risk CAD.

Limitations of stress echocardiography

Although the current study suggests that stress echocardiography has better discriminatory capacity for the detection of high-risk CAD, this technique has its own inherent limitations.⁷⁸ False negativity has been observed in patients who do not achieve target workload.⁷⁹ The lack of interpretable images during exercise (obese, chronic obstructive pulmonary disease), delay in recording images post treadmill (secondary to intense respiratory effort and discomfort following exercise) and inability to image the entire left ventricle in some patients can adversely affect interpretation. Lack of quantitation and visual assessment essentially contribute to the concerning inter-observer and inter-institutional variability.⁸⁰ Small LV size can make interpretation of RWMA more challenging.⁸ The hallmark of ischemia (per cent LV wall thickening) may be limited by increased diastolic thickness (compensatory response to decreased LV preload from dobutamine) as compared to LV systolic thickness.⁸¹ Conversely, localised basal inferior wall abnormalities, cardiomyopathies and abnormal septal wall motion (post bypass, left bundle branch block) can lead to false positive results.^{31–32} Dobutamine infusion may also be associated with ventricular arrhythmias in patients with low ejection fraction and high-risk CAD.

Referral (verification) bias

It is plausible that referral bias contributes to an overestimation of the sensitivity of MPI in the detection of high-risk CAD as the extent of ischemia is generally the primary determinant of the need for subsequent catheterisation and/or revascularisation. Notably, a significant proportion of patients with significant LM disease or multivessel disease may not undergo cardiac catheterisation because of reportedly normal perfusion scans. This issue remains poorly defined and the precise magnitude of underestimation of high-risk CAD remains obscure due to the paucity of reported data on patients who have concurrently undergone both stress testing and angiography to enable a direct comparison.

Although it is unclear what threshold of high-risk anatomy is predictive of the temporal propensity for a cardiac event, in large cohorts, a normal nuclear perfusion scan, even in patients with a high pretest likelihood of CAD, is associated with a low annual event rate (cardiac death and hard event rate of 0.6% and 1.3%, respectively)⁸³ and when detected, the extent of ischemia has been shown to correlate with short-term survival benefit from revascularisation.^{84–85} Similarly, a normal exercise echocardiogram result is associated with a <1% annual event rate of cardiac death and nonfatal myocardial infarction, comparable to that of an age and sex-matched population.^{86–88}

Clinical implications

For the overall detection of CAD, two meta-analyses comparing exercise SE and exercise MPI showed similar sensitivities for the

detection of CAD for both modalities but higher specificities and discriminatory capabilities for SE.^{89–90}

Both stress modalities have been validated extensively for assessing prognosis, with a negative test result associated with a low “hard event” rate at follow-up.^{18–63–64–83–86–88} Based on recent meta-analytic data, the negative predictive value over approximately 3-year follow-up for both techniques is high, reported to be 98.8% and 98.4% for exercise MPI and exercise echocardiography, respectively, with equivalent prognostic utility in both women and men.⁹¹ Another analysis reported an annual cardiac event rate of <1.2% and an annual cardiac death rate of 0.08% for both stress modalities.⁹ Similar data have been reported for perioperative risk stratification and outcomes,^{2–92} with one meta analysis of patients undergoing major vascular surgery suggesting a trend for better diagnostic accuracy with SE.⁹³

The current study presents a meta-analysis of the available literature on SE and MPI for the detection of LM and TVD. In these high-risk subsets of CAD, our analysis suggests that SE may be superior to MPI for screening of patients for LM and TVD.

Although SE has been shown to be sensitive for diagnosing CAD, it is less useful for the accurate localisation of the diseased vessels, particularly in the setting of LM and TVD compared to single-vessel disease.⁹⁴ This may be due to the fact that this modality relies on the identification of a combination of symptoms, hemodynamic parameters and more importantly wall motion abnormalities. Additionally, disparate ischemic thresholds in areas subtended by functionally significant stenoses in patients with TVD may result in the rapid recognition of contractile abnormalities with or without ECG symptoms or chest pain in the territory subtended by the most significant stenosis, typically prompting the termination of the test and precluding the detection of ischemia in segments with higher ischemic thresholds. The lack of radiation, relative accuracy, availability and practicality of SE versus MPI are also important considerations that have direct impact and implications for screening CAD programs and chest pain centres.

Limitations

First, there are inherent drawbacks to comparing non-invasive tests that address the physiologic significance of a stenosis to coronary anatomy defined by angiography with its well-described shortcomings.⁹⁵ Nevertheless, invasive coronary angiography remains the gold standard for the diagnosis and assessment of extent and severity of CAD. Second, heterogeneity between study design and results is a common problem in any systematic review or meta-analysis for diagnostic tests. Identifying its presence, investigating its cause and correctly accounting for it are important aspects to be addressed by every researcher. Such a comparison has to ideally be adjusted for study and patient characteristics to reduce the variation between studies. There is published evidence that the overall validity of the adjusted indirect comparisons depends on the internal validity and similarity of the included studies.^{23–25} Hence, we use weighted analysis (pooled sensitivity, specificity, likelihood ratios weighted by the inverse of variance of sensitivity, specificity and likelihood ratios, meta-regression analysis weighted by the inverse of variance of DOR in each study) using a random effects model. The presence of heterogeneity in the included studies (resulting from threshold effect, different study population, differences in technology, operators and study designs) may affect the overall validity of the pooled estimates. However, the SROC analysis is robust to heterogeneity.

Cardiac imaging and non-invasive testing

Additionally, we explored data using meta-regression analysis with the following covariates (mean age, publication year, sex).

Meta analysis is also subject to publication bias—that is, studies with a significant result are more likely to be submitted and published. In order to minimise this bias, we did a comprehensive screening of both full-text articles and abstracts (with unpublished manuscripts). However, only those articles that used coronary angiography as the reference standard and where data was abstractable for patients with left main and triple vessel disease were included in this analysis. It was ruled out using precision-weighted linear regression.²⁸ Additionally, we tested the effect of potential modifiers such as year of publication, mean age, sex (proportion of men), clinical indication and type of stressor on test performance in multivariate analysis (significance level of $p < 0.05$). The protocols for conducting stress tests (including holding certain medications), tracers and methods of interpretation and reporting (with inherent inter observer variations) vary from one institution to another and may affect the results. The information on the assessment of wall motion with low dose stressor and the classic biphasic response with low- and high-dose stressor was not consistently available in studies with SE. None of the studies with SE used recovery phase wall motion assessments. The low specificities for both the stress modalities in our study are secondary to the way disease was defined. Patients without LM or TVD were defined as not diseased. However, this procedure was uniformly followed for both SE and MPI to avoid bias. Additionally, the absence of quantitative coronary angiography may have introduced observer bias in the identification of high-risk CAD. The specificity of MPI and SE was similar in our study in contrast to published literature that shows higher specificity of SE for overall detection of CAD.⁸⁹

In patients with reduced left ventricular function, the issue of identification of patients with left main and triple vessel disease (vs. global hypokinesis in non-ischaemic, primary dilated cardiomyopathy) is clinically relevant. However, with the exception of one study,³¹ the absence of uniformly abstractable data on high-risk CAD patients with reduced ejection fraction from studies included in this meta-analysis thwarted attempts to examine this important question further. Use of computed tomographic angiography or hybrid imaging (to quantitate absolute blood flow and assess the physiologic significance of stenotic lesions) may help circumvent the pitfalls of MPI and SE in this subset of patients.

Finally, the limitations of this study also include the limitations of indirect meta-regressive techniques. Results of indirect comparisons do not always agree with the results of direct comparative studies, even randomised control trials.²³ A contemporary prospective head to head comparison of the two imaging modalities can overcome this limitation. In the absence of direct comparative data published or available at the present time, these findings represent the only systematic analysis of the literature on this clinically relevant issue.

CONCLUSIONS

Our meta-analysis suggests that SE has superior overall discriminatory capacity (SROC analysis) with greater sensitivity and lower negative LR compared to myocardial perfusion imaging for the detection of LM or TVD. Our findings appear to indicate that SE may be less vulnerable than myocardial perfusion imaging to false negative study results when performed in persons with high-risk CAD.

Acknowledgements There are no potential conflicts of interest to be disclosed by all the authors. All the authors had access to the data and made substantial contributions to the work reported in this manuscript.

Competing interests None.

Patient consent Obtained.

Provenance and peer review Not commissioned; externally peer reviewed.

REFERENCES

- Gibbons RJ, Abrams J, Chatterjee K, *et al.* ACC/AHA 2002 guideline update for the management of patients with chronic stable angina—summary article: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines (committee on the management of patients with chronic stable angina). *J Am Coll Cardiol* 2003;**41**:159–68.
- Shaw LJ, Eagle KA, Gersh BJ, *et al.* Meta-analysis of intravenous dipyridamole-thallium-201 imaging (1985 to 1994) and dobutamine echocardiography (1991 to 1994) for risk stratification before vascular surgery. *J Am Coll Cardiol* 1996;**27**:787–98.
- Hoffmann R, Lethen H, Kuhl H, *et al.* Extent and severity of test positivity during dobutamine stress echocardiography. Influence on the predictive value for coronary artery disease. *Eur Heart J* 1999;**20**:1485–92.
- Picano E, Bedetti G, Varga A, *et al.* The comparable diagnostic accuracies of dobutamine stress and dipyridamole-stress echocardiographies: a meta-analysis. *Coron Artery Dis* 2000;**11**:151–9.
- Takeuchi M, Sonoda S, Miura Y, *et al.* Comparative diagnostic value of dobutamine stress echocardiography and stress thallium-201 single-photon-emission computed tomography for detecting coronary artery disease in women. *Coron Artery Dis* 1996;**7**:831–5.
- Ho FM, Huang PJ, Liao CS, *et al.* Dobutamine stress echocardiography compared with dipyridamole thallium-201 single-photon emission computed tomography in detecting coronary artery disease. *Eur Heart J* 1995;**16**:570–5.
- Takeuchi M, Araki M, Nakashima Y, *et al.* Comparison of dobutamine stress echocardiography and stress thallium-201 single-photon emission computed tomography for detecting coronary artery disease. *J Am Soc Echocardiogr* 1993;**6**:593–602.
- Smart SC, Knickelbine T, Malik F, *et al.* Dobutamine-atropine stress echocardiography for the detection of coronary artery disease in patients with left ventricular hypertrophy. Importance of chamber size and systolic wall stress. *Circulation* 2000;**101**:258–63.
- Schinkel AF, Bax JJ, Geleijnse ML, *et al.* Noninvasive evaluation of ischaemic heart disease: myocardial perfusion imaging or stress echocardiography? *Eur Heart J* 2003;**24**:789–800.
- Noto TJ Jr, Johnson LW, Krone R, *et al.* Cardiac catheterization 1990: a report of the Registry of the Society for Cardiac Angiography and Interventions (SCAI). *Cathet Cardiovasc Diagn* 1991;**24**:75–83.
- Cohen MV, Gorlin R. Main left coronary artery disease. Clinical experience from 1964–74. *Circulation* 1975;**52**:275–85.
- Bruschke AV, Proudfit WL, Sones FM Jr. Progress study of 590 consecutive nonsurgical cases of coronary disease followed 5–9 years. II. Ventriculographic and other correlations. *Circulation* 1973;**47**:1154–63.
- Califf RM, Armstrong PW, Carver JR, *et al.* 27th Bethesda conference: matching the intensity of risk factor management with the hazard for coronary disease events. Task Force 5. Stratification of patients into high, medium and low risk subgroups for purposes of risk factor management. *J Am Coll Cardiol* 1996;**27**:1007–19.
- Campeau L, Corbara F, Crochet D, *et al.* Left main coronary artery stenosis: the influence of aortocoronary bypass surgery on survival. *Circulation* 1978;**57**:1111–15.
- McConahay DR, Killen DA, McCallister BD, *et al.* Coronary artery bypass surgery for left main coronary artery disease. *Am J Cardiol* 1976;**37**:885–9.
- Killen DA, Reed WA, Kindred L, *et al.* Surgical therapy for left main coronary artery disease. *J Thorac Cardiovasc Surg* 1980;**80**:255–61.
- Yusuf S, Zucker D, Peduzzi P, *et al.* Effect of coronary artery bypass graft surgery on survival: overview of 10-year results from randomised trials by the Coronary Artery Bypass Graft Surgery Trialists Collaboration. *Lancet* 1994;**344**:563–70.
- Hachamovitch R, Berman DS, Kiat H, *et al.* Exercise myocardial perfusion SPECT in patients without known coronary artery disease: incremental prognostic value and use in risk stratification. *Circulation* 1996;**93**:905–14.
- Mairesse GH, Marwick TH, Vanoverschelde JL, *et al.* How accurate is dobutamine stress electrocardiography for detection of coronary artery disease? Comparison with two-dimensional echocardiography and technetium-99m methoxy isobutyl isonitrile (mibi) perfusion scintigraphy. *J Am Coll Cardiol* 1994;**24**:920–7.
- Lusted LB. Signal detectability and medical decision-making. *Science* 1971;**171**:1217–19.
- Walter SD. Small sample estimation of log odds ratios from logistic regression and fourfold tables. *Stat Med* 1985;**4**:437–44.
- Walter SD. Properties of the summary receiver operating characteristic (SROC) curve for diagnostic test data. *Stat Med* 2002;**21**:1237–56.
- Song F, Altman DG, Glenny AM, *et al.* Validity of indirect comparison for estimating efficacy of competing interventions: empirical evidence from published meta-analyses. *BMJ* 2003;**326**:472.
- Glas AS, Lijmer JG, Prins MH, *et al.* The diagnostic odds ratio: a single indicator of test performance. *J Clin Epidemiol* 2003;**56**:1129–35.
- Bucher HC, Guyatt GH, Griffith LE, *et al.* The results of direct and indirect treatment comparisons in meta-analysis of randomized controlled trials. *J Clin Epidemiol* 1997;**50**:683–91.

26. **Higgins JP**, Thompson SG, Deeks JJ, *et al.* Measuring inconsistency in meta-analyses. *BMJ* 2003;**327**:557–60.
27. **Lau J**, Ioannidis JP, Schmid CH. Quantitative synthesis in systematic reviews. *Ann Intern Med* 1997;**127**:820–6.
28. **Egger M**, Davey Smith G, Schneider M, *et al.* Bias in meta-analysis detected by a simple, graphical test. *BMJ* 1997;**315**:629–34.
29. **Jaeschke R**, Goyatt G, Sackett DL. Users' guides to the medical literature. III. How to use an article about a diagnostic test. A. Are the results of the study valid? Evidence-based medicine working group. *JAMA* 1994;**271**:389–91.
30. **Moses LE**, Shapiro D, Littenberg B. Combining independent studies of a diagnostic test into a summary ROC curve: data-analytic approaches and some additional considerations. *Stat Med* 1993;**12**:1293–316.
31. **Sharp SM**, Sawada SG, Segar DS, *et al.* Dobutamine stress echocardiography: detection of coronary artery disease in patients with dilated cardiomyopathy. *J Am Coll Cardiol* 1994;**24**:934–9.
32. **Andrade MJ**, Picano E, Pingitore A, *et al.* Dipyridamole stress echocardiography in patients with severe left main coronary artery narrowing. Echo Persantine International Cooperative (EPIC) study group—subproject "left main detection". *Am J Cardiol* 1994;**73**:450–5.
33. **Berman DS**, Kang X, Slomka PJ, *et al.* Underestimation of extent of ischemia by gated SPECT myocardial perfusion imaging in patients with left main coronary artery disease. *J Nucl Cardiol* 2007;**14**:521–8.
34. **Bjornstad K**, Aakhus S, Hatle L. Comparison of digital dipyridamole stress echocardiography and upright bicycle stress echocardiography for identification of coronary artery stenosis. *Cardiology* 1995;**86**:514–20.
35. **Chikamori T**, Doi YL, Yonezawa Y, *et al.* Noninvasive identification of significant narrowing of the left main coronary artery by dipyridamole thallium scintigraphy. *Am J Cardiol* 1991;**68**:472–7.
36. **Elhendy A**, Geleijnse ML, van Domburg RT, *et al.* Comparison of dobutamine stress echocardiography and technetium 99m sestamibi single photon emission tomography for the diagnosis of coronary artery disease in hypertensive patients with and without left ventricular hypertrophy. *Eur J Nucl Med* 1998;**25**:69–78.
37. **Elhendy A**, van Domburg RT, Bax JJ, *et al.* Noninvasive diagnosis of coronary artery stenosis in women with limited exercise capacity: comparison of dobutamine stress echocardiography and 99mTc sestamibi single-photon emission CT. *Chest* 1998;**114**:1097–104.
38. **Epstein M**, Gin K, Sterns L, *et al.* Dobutamine stress echocardiography: initial experience of a Canadian centre. *Can J Cardiol* 1992;**8**:273–9.
39. **Fragasso G**, Lu C, Dabrowski P, *et al.* Comparison of stress/rest myocardial perfusion tomography, dipyridamole and dobutamine stress echocardiography for the detection of coronary disease in hypertensive patients with chest pain and positive exercise test. *J Am Coll Cardiol* 1999;**34**:441–7.
40. **Hecht HS**, DeBord L, Shaw R, *et al.* Supine bicycle stress echocardiography versus tomographic thallium-201 exercise imaging for the detection of coronary artery disease. *J Am Soc Echocardiogr* 1993;**6**:177–85.
41. **Hennessy TG**, Codd MB, McCarthy C, *et al.* Dobutamine stress echocardiography in the detection of coronary artery disease in a clinical practice setting. *Int J Cardiol* 1997;**62**:55–62.
42. **Kurata C**, Tawaraha K, Taguchi T, *et al.* Lung thallium-201 uptake during exercise emission computed tomography. *J Nucl Med* 1991;**32**:417–23.
43. **Solot G**, Hermans J, Merlo P, *et al.* Correlation of 99Tcm-sestamibi SPECT with coronary angiography in general hospital practice. *Nucl Med Communications* 1993;**14**:23–9.
44. **Maddahi J**, Abdulla A, Garcia EV, *et al.* Noninvasive identification of left main and triple vessel coronary artery disease: improved accuracy using quantitative analysis of regional myocardial stress distribution and washout of thallium-201. *J Am Coll Cardiol* 1986;**7**:53–60.
45. **Mahmarijan JJ**, Boyce TM, Goldberg RK, *et al.* Quantitative Exercise thallium-201 single photon emission computed tomography for the enhanced diagnosis of ischemic heart disease. *J Am Coll Cardiol* 1990;**15**:318–29.
46. **Nygaard TW**, Gibson RS, Ryan JM, *et al.* Prevalence of high-risk thallium-201 scintigraphic findings in left main coronary artery stenosis: comparison with patients with multiple- and single-vessel coronary artery disease. *Am J Cardiol* 1984;**53**:462–9.
47. **Oguzhan A**, Kisacik HL, Ozdemir K, *et al.* Comparison of exercise stress testing with dobutamine stress echocardiography and exercise technetium-99m isonitrite single photon emission computed tomography for diagnosis of coronary artery disease. *Jpn Heart J* 1997;**38**:333–44.
48. **Prince CR**, Stoddard MF, Morris GT, *et al.* Dobutamine two-dimensional transesophageal echocardiographic stress testing for detection of coronary artery disease. *Am Heart J* 1994;**128**:36–41.
49. **Quiñones MA**, Verani MS, Haichin RM, *et al.* Exercise echocardiography versus 201Tl single photon emission computed tomography in evaluation of coronary artery disease: analysis of 292 patients. *Circulation* 1992;**85**:1026–31.
50. **Rehn T**, Griffith LS, Achuff SC, *et al.* Exercise thallium-201 myocardial imaging in left main coronary artery disease: sensitive but not specific. *Am J Cardiol* 1981;**48**:217–23.
51. **Ryan T**, Segar DS, Sawada SG, *et al.* Detection of coronary artery disease with upright bicycle exercise echocardiography. *J Am Soc Echocardiogr* 1993;**6**:186–97.
52. **Sawada SG**, Segar DS, Ryan T, *et al.* Echocardiographic detection of coronary artery disease during dobutamine infusion. *Circulation* 1991;**83**:1605–14.
53. **Severi S**, Picano E, Michelassi C, *et al.* Diagnostic and prognostic value of dipyridamole echocardiography in patients with suspected coronary artery disease. Comparison with exercise electrocardiography. *Circulation* 1994;**89**:1160–73.
54. **Agati L**, Renzi M, Sciomer S, *et al.* Transesophageal dipyridamole echocardiography for diagnosis of coronary artery disease. *J Am Coll Cardiol* 1992;**19**:765–70.
55. **Lambert H**, Kreis A, Trümper H, *et al.* Simultaneous transesophageal atrial pacing and transesophageal two-dimensional echocardiography: a new method of stress echocardiography. *J Am Coll Cardiol* 1990;**16**:1143–53.
56. **Takeishi Y**, Chiba J, Abe S, *et al.* Noninvasive identification of left main and three vessel coronary artery disease by thallium-201 single photon emission computed tomography during adenosine infusion. *Ann Nucl Med* 1994;**8**:1–7.
57. **Wakasugi S**, Shibata N, Kobayashi T, *et al.* Specific perfusion pattern in stress 201Tl myocardial scintigraphy of left main coronary artery disease. *Eur J Nucl Med* 1986;**12**:369–74.
58. **Zaacks SM**, Ali A, Parrillo JE, *et al.* How well does radionuclide dipyridamole stress testing detect three-vessel coronary artery disease and ischemia in the region supplied by the most stenotic vessel? *Clin Nucl Med* 1999;**24**:35–41.
59. **Williams KA**, Schneider CM. Increased stress right ventricular activity on dual isotope perfusion SPECT: A sign of multivessel and/or left main coronary artery disease. *J Am Coll Cardiol* 1993;**34**:420–7.
60. **Lima RSL**, Watson DD, Goode AR, *et al.* Incremental value of combined perfusion and function over perfusion alone for gated SPECT myocardial perfusion imaging for detection of severe three-vessel coronary artery disease. *J Am Coll Cardiol* 2003;**42**:64–70.
61. **Afonso L**, Mahajan N. Single photon emission computed tomography myocardial perfusion imaging in the diagnosis of left main disease. *Clin Cardiol* 2009;**32**:E11–15.
62. **Rallidis L**, Cokkinos P, Tousoulis D, *et al.* Comparison of dobutamine and treadmill exercise echocardiography in inducing ischemia in patients with coronary artery disease. *J Am Coll Cardiol* 1997;**30**:1660–8.
63. **Geleijnse ML**, Elhendy A, van Domburg RT, *et al.* Cardiac imaging for risk stratification with dobutamine-atropine stress testing in patients with chest pain echocardiography, perfusion scintigraphy, or both? *Circulation* 1997;**96**:137–47.
64. **Olmos LJ**, Dakik H, Gordon R, *et al.* Long-term prognostic value of exercise echocardiography compared with exercise 201Tl, ECG, and clinical variables in patients evaluated for coronary artery disease. *Circulation* 1998;**98**:2679–86.
65. **Nesto RW**, Kowalchuk GJ. The ischemic cascade: temporal sequence of hemodynamic, electrocardiographic and symptomatic expressions of ischemia. *Am J Cardiol* 1987;**59**:23C–30C.
66. **Travin M**. Is it possible for myocardial perfusion imaging to avoid missing any patients with high-risk coronary disease? *J Nucl Cardiol* 2007;**14**:492–96.
67. **Aarnoudse WH**, Botman KJ, Pijls NH. False-negative myocardial scintigraphy in balanced three-vessel disease revealed by coronary pressure measurement. *Int J Cardiovasc Intervent* 2003;**5**:67–71.
68. **Ragosta M**, Bishop AH, Lipson LC, *et al.* Comparison between angiography and fractional flow reserve versus single-photon emission computed tomographic myocardial perfusion imaging for determining lesion significance in patients with multivessel coronary disease. *Am J Cardiol* 2007;**99**:896–902.
69. **Abidov A**, Hachamovitch R, Hayes SW, *et al.* Prognostic impact of hemodynamic response to adenosine in patients older than age 55 years undergoing vasodilator stress myocardial perfusion study. *Circulation* 2003;**107**:2894–9.
70. **Duvernoy CS**, Ficaro EP, Karabadjian MZ, *et al.* Improved detection of left main coronary artery disease with attenuation corrected SPECT. *J Nucl Cardiol* 2000;**7**:639–48.
71. **Romanens M**, Gradel C, Saner H, *et al.* Comparison of 99mTc-sestamibi lung/heart ratio, transient ischaemic dilation and perfusion defect size for the identification of severe and extensive coronary artery disease. *Eur J Nucl Med* 2001;**28**:907–10.
72. **Emmett L**, Magee M, Freedman SB, *et al.* The role of left ventricular hypertrophy and diabetes in the presence of transient ischemic dilation of the left ventricle on myocardial perfusion SPECT images. *J Nucl Med* 2005;**46**:1596–601.
73. **Kwok JM**, Christian TF, Miller TD, *et al.* Identification of severe coronary artery disease in patients with a single abnormal coronary territory on exercise thallium-201 imaging: the importance of clinical and exercise variables. *J Am Coll Cardiol* 2000;**35**:335–44.
74. **Williams KA**, Schuster RA, Williams KA Jr, *et al.* Correct spatial normalization of myocardial perfusion SPECT improves detection of multivessel coronary artery disease. *J Nucl Cardiol* 2003;**10**:353–60.
75. **Groch SW**, Schippers DJ, Marshall RC, *et al.* Quantitative gated blood pool SPECT: analysis of 3-dimensional models for the assessment of regional myocardial wall motion. *J Nucl Cardiol* 2002;**9**:271–84.
76. **Dorbala S**, Vangala D, Sampson U, *et al.* Value of vasodilator left ventricular ejection fraction reserve in evaluating the magnitude of myocardium at risk and the extent of angiographic coronary artery disease: a 82Rb PET/CT study. *J Nucl Med* 2007;**48**:349–58.
77. **Sampson UK**, Dorbala S, Limaye A, *et al.* Diagnostic accuracy of rubidium-82 myocardial perfusion imaging with hybrid positron emission tomography/computed tomography in the detection of coronary artery disease. *J Am Coll Cardiol* 2007;**49**:1052–8.
78. **Pellikka PA**, Nagueh SF, Elhendy AA, *et al.* American Society of Echocardiography recommendations for performance, interpretation, and application of stress echocardiography. *J Am Soc Echocardiogr* 2007;**20**:1021–41.
79. **Ling LH**, Pellikka PA, Mahoney DW, *et al.* Atropine augmentation in dobutamine stress echocardiography: role and incremental value in a clinical practice setting. *J Am Coll Cardiol* 1996;**28**:551–7.
80. **Hoffmann R**, Lethen H, Marwick T, *et al.* Analysis of interinstitutional observer agreement in interpretation of dobutamine stress echocardiograms. *J Am Coll Cardiol* 1996;**27**:330–6.

Cardiac imaging and non-invasive testing

81. **Hausnerova E**, Gottdiener J, Hecht G, *et al*. Increased diastolic left ventricular wall thickness during dobutamine stress echocardiography: effect of afterload. *Circulation* 1994;**90**:391 (abstract).
82. **MarwickTH**. Stress Echocardiography. *Heart* 2003;**89**:113–18.
83. **Hachamovitch R**, Hayes SW, Friedman JD, *et al*. Stress myocardial perfusion single-photon emission computed tomography is clinically effective and cost effective in risk stratification of patients with a high likelihood of coronary artery disease (CAD) but no known CAD. *J Am Coll Cardiol* 2004;**43**:200–8.
84. **Hachamovitch R**, Rozanski A, Hayes SW, *et al*. Predicting therapeutic benefit from myocardial revascularization procedures: are measurements of both resting left ventricular ejection fraction and stress-induced myocardial ischemia necessary? *J Nucl Cardiol* 2006;**13**:768–78.
85. **Hachamovitch R**, Hayes SW, Friedman JD, *et al*. Comparison of the short-term survival benefit associated with revascularization compared with medical therapy in patients with no prior coronary artery disease undergoing stress myocardial perfusion single photon emission computed tomography. *Circulation* 2003;**107**:2900–7.
86. **Sawada SG**, Ryan T, Conley MJ, *et al*. Prognostic value of a normal exercise echocardiogram. *Am Heart J* 1990;**120**:49–55.
87. **Elhendy A**, Mahoney DW, McCully RB, *et al*. Use of a scoring model combining clinical, exercise test, and echocardiographic data to predict mortality in patients with known or suspected coronary artery disease. *Am J Cardiol* 2004;**93**:1223–8.
88. **Elhendy A**, Mahoney DW, Burger KN, *et al*. Prognostic value of exercise echocardiography in patients with classic angina pectoris. *Am J Cardiol* 2004;**94**:559–63.
89. **Fleischmann KE**, Hunink MG, Kuntz KM, *et al*. Exercise echocardiography or exercise SPECT imaging? A meta-analysis of diagnostic test performance. *JAMA* 1998;**280**:913–20.
90. **Heijenbrok-Kal MH**, Fleischmann KE, Hunink MG. Stress echocardiography, stress single-photon emission computed tomography and electron beam computed tomography for the assessment of coronary artery disease: a meta-analysis of diagnostic performance. *Am Heart J* 2007;**154**:415–23.
91. **Metz LD**, Beattie M, Hom R, *et al*. The prognostic value of normal exercise myocardial perfusion imaging and exercise echocardiography: a meta-analysis. *J Am Coll Cardiol* 2007;**49**:227–37.
92. **Kontos MC**, Akosah KO, Brath LK, *et al*. Cardiac complications in noncardiac surgery: value of dobutamine stress echocardiography versus dipyridamole thallium imaging. *J Cardiothorac Vasc Anesth* 1996;**10**:329–35.
93. **Kertai MD**, Boersma E, Bax JJ, *et al*. A meta-analysis comparing the prognostic accuracy of six diagnostic tests for predicting perioperative cardiac risk in patients undergoing major vascular surgery. *Heart* 2003;**89**:1327–34.
94. **Ho YL**, Wu CC, Chao CL, *et al*. Localizing individual coronary artery obstructions with the dobutamine stress echocardiography. *Cardiology* 1997;**88**:197–202.
95. **White CW**, Wright CB, Doty DB, *et al*. Does visual interpretation of the coronary arteriogram predict the physiologic importance of a coronary stenosis? *N Engl J Med* 1984;**310**:819–24.

Heart

Diagnostic accuracy of myocardial perfusion imaging and stress echocardiography for the diagnosis of left main and triple vessel coronary artery disease: a comparative meta-analysis

N Mahajan, L Polavaram, H Vankayala, B Ference, Y Wang, J Ager, J Kovach and L Afonso

Heart 2010 96: 956-966
doi: 10.1136/hrt.2009.182295

Updated information and services can be found at:
<http://heart.bmj.com/content/96/12/956>

	<i>These include:</i>
References	This article cites 95 articles, 25 of which you can access for free at: http://heart.bmj.com/content/96/12/956#BIBL
Email alerting service	Receive free email alerts when new articles cite this article. Sign up in the box at the top right corner of the online article.

Topic Collections

Articles on similar topics can be found in the following collections

[Drugs: cardiovascular system](#) (8135)
[Clinical diagnostic tests](#) (4480)
[Echocardiography](#) (1963)
[Epidemiology](#) (3362)

Notes

To request permissions go to:
<http://group.bmj.com/group/rights-licensing/permissions>

To order reprints go to:
<http://journals.bmj.com/cgi/reprintform>

To subscribe to BMJ go to:
<http://group.bmj.com/subscribe/>